

Zian(Andy) Zheng

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Education

MMath University of Waterloo , Computer Science, Ontario, Canada	Sept 2024 – Now
MComp National University of Singapore , Artificial Intelligence, Singapore	Sept 2022 – May 2024
Advised by Presidential Young Professor Yang You	
BEng Lanzhou University , Data Science, Lanzhou, China	Sept 2018 – May 2022
GPA: 92.8/100 (Ranking: 1/192)	

Research Projects

OpenMoE: An Early Effort on Open Mixture-of-Experts Language Models - Worked on the Pytorch implementation of the first open-source, decoder-only MoE language model, OpenMoE, providing insights about the routing mechanism to the open-source community [model checkpoint]. - Prepared the training dataset, tokenizer and conducted the model evaluation.	Github Repo
Matrix: Infinite-Horizon World Generation with Real-Time Interaction - Built a data collection pipeline for Cyberpunk 2077, recording per-frame video data and corresponding control signals to support model training - Served as a core contributor to a multi-GPU inference framework, implementing Ray-based workers (DiT, VAE, post-processing) and building an interactive front-end/backend system. Delivered the first real-time game generation demo with user-controllable inference [demo].	Github Repo
AdaVocab: Boosting SLM Inference with Sparse Vocabulary Activation - Identified the growing vocabulary size as a major bottleneck for the Small Language Model (SLM) inference efficiency. - Proposed and implemented a sparsely active vocabulary method; prepared training data, modeled Trainer, and completed evaluation with teammates. - Achieved over 20% computation reduction and 10% inference speedup for SLMs in CPU settings.	Github Repo

Work Experience

HPC-AI Tech , Artificial Intelligence Engineer Intern	Beijing, China July 2023 – Nov 2023
Extended LLaMA's vocabulary for Chinese and contributed to data preparation in the Colossal-LLaMA-2 project, selected as an official base model in the 2023 NeurIPS LLM Efficiency Challenge. - Investigated common context length extrapolation methods (e.g. PI, NTK, LongLoRA), and implemented corresponding training and evaluation pipelines to extrapolate Colossal-LLaMA-2 with multi-GPU training. - Working on the ColossalQA project, a RAG framework based on Langchain.	

Publications

[ICML 2024] OpenMoE: An Early Effort on Open Mixture-of-Experts Language Models	Jan 2024
Fuzhao Xue, Zian Zheng , Yao Fu, Jinjie Ni, Zangwei Zheng, Wangchunshu Zhou, Yang You	

[ICLR 2025] [🔗](#)MixEval-X: Any-to-Any Evaluations from Real-World Data Mixtures

Oct 2024

Jinjie Ni, Yifan Song, Deepanway Ghosal, Bo Li, David Junhao Zhang, Xiang Yue, Fuzhao Xue,
Zian Zheng, Kaichen Zhang, Mahir Shah, Kabir Jain, Yang You, Michael Shieh

[Arxiv] [🔗](#)The Matrix: Infinite-Horizon World Generation with Real-Time Moving Control

Dec 2024

Ruili Feng, Han Zhang, Zhantao Yang, Jie Xiao, Zhilei Shu, Zhiheng Liu, **Andy Zheng**, Yukun Huang,
Yu Liu, Hongyang Zhang

Teaching & Service

Teaching Assistant: CS135, CS479/679 at UWaterloo

Reviewers: ICLR 2025, AISTATS 2025

Honors & Awards

China National Scholarship (Top 0.1% across nation), 12/2019 & 12/2021

Merit Student in Colleges and Universities in Gansu Province (Top 1% across province), 05/2021

Dr. Derick Wood Graduate Scholarship, 12/2024

Computer Skills

Programming & Software: Python, Java, C, SQL, Tableau, Echarts, Linux, Hadoop

Libraries: PyTorch, Ray, Pandas, NumPy, Scikit-Learn, PyQt5